Median
10%- / 90%-Fraktil

Budget: 100 kgCO_2 pro m Gurtlänge

N_{Rd} [MN]

Zuggurt

Vollholz C24

Brettschichtholz GL24h

Baustahl S355

50 mm

100 mm

40 mm

14 mm

Stahlbeton C30/37, vorgespannt (10 MPa) ρ_p = 0.9%

17.5 15 12.5 10 7.5 5 2.5 -5 -7.5 -10 -12.5 -15 -17.5 -20 -22.5 -25

-4.0 -3.5 -3.0 -2.5 -2.0 -1.5 -1.0 -0.5 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0

u [mm]

100 mm

50 mm

32 mm

Betonstahl B500B (nackt)

Stahlbeton C30/37, ρ_s=0.6%

Umschnürter Beton C30/37 ρ_w=8.6%, ρ_l=0.6%

Druckgurt

Stahlbeton C30/37
Zug: ρ_s = 10%
A = 159'261 mm^2
g_{0k} = 4.7 kN/m
Druck: ρ_s = 0.6%
A = 430'563 mm^2
g_{0k} = 10.5 kN/m

Umschnürter Beton
ρ_w = 8.6%, ρ_l = 0.6%
A = 162'767 mm^2
g_{0k} = 5.1 kN/m

Brettschichtholz
GL24h
A = 1'000'000 mm^2
g_{0k} = 4.4 kN/m

Vollholz
C24
A = 1'818'182 mm^2
g_{0k} = 8.6 kN/m

vorgespannter Beton
σ_{cp} = 10 MPa
ρ_p = 0.9%
A_p = 3'239 mm^2
A_c = 358'461 mm^2
g_{0k} = 8.9 kN/m

Baustahl
A = 20'730 mm^2
g_{0k} = 1.6 kN/m

Betonstahl
A = 22'645 mm^2
g_{0k} = 1.7 kN/m